

JASON FORCE

Perception • Robotics • Systems Engineering

jason@posechain.net | github.com/posechain

Northern Virginia | (703) 217-2027

Professional Summary

Multi-disciplinary perception, robotics, and electromechanical systems engineer with expertise spanning computer vision, navigation, firmware, controls, optomechanical systems, and heterogeneous computing architectures. Experienced developing and integrating sensing, software, hardware, and autonomy-related systems for robotics, UAV, aerospace, and advanced technical applications. Background includes perception systems, precision motion and pointing mechanisms, optics and laser-based systems, and multidisciplinary system integration for complex technical programs.

Core Technical Expertise

Robotics & Navigation: ROS 2, VINS-Fusion, PX4, MAVROS, MAVLink, Gazebo, GTSAM with ISAM, SLAM, autonomous navigation

Tracking & Perception: Computer vision target tracking, motion estimation, visual tracking

Computer Vision & AI: PyTorch, CNNs, OpenCV, Open3D, Scikit-learn, machine learning

Controls & Estimation: Kalman filtering, factor graphs, state estimation, sensor fusion, feedback control

Software & Firmware: Python, C++, firmware, Embedded Linux, RTOS, ARM architectures, asynchronous systems, real-time control

Optics, Motion & Mechatronics: Optomechanical design, precision motion systems, frameless motors, motor drive systems, high-power laser systems, precision bearings, pointing systems

Hardware & Processing: CPU/GPU/TPU/DSP/FPGA architectures, microprocessors, AI coprocessors, Intel RealSense, Ambarella, Hailo

Engineering & Development Tools: Git, VSCode, PyCharm, Jupyter Notebooks, Google Colab, Amazon SageMaker, Claude, Siemens NX

Technical Communications & Planning: Technical documentation, system architecture communication, Confluence, Jira, EaseRequirements, requirements traceability, multidisciplinary technical coordination

Professional Experience

Software Lead

Overview Energy | Sep 2024 – Present

- Serve as technical lead supporting spacecraft software, subsystem design, and cross-domain technical development for space-based energy systems.
- Participate in system-level architecture trade studies and interface analysis for spacecraft bus-to-payload communications, including CCSDS and MaxPackets approaches.
- Support spacecraft subsystem design involving high-power laser power-beaming architectures, optics, controls, and precision electromechanical systems.
- Perform optomechanical and mechanical design using Siemens NX, supporting precision motion system development including frameless motors, precision bearings, and pointing-related mechanisms.
- Develop firmware and software interfaces for spacecraft-related subsystems using serial communication protocols including CAN, SPI, I2C, and SSI encoder interfaces.

Mobile Robotics Perception Engineer

Force Consulting Services LLC | Dec 2023 – Sept 2024

- Developed and evaluated visual inertial navigation system (VINS) approaches for UAV navigation in varied operating environments.

- Developed geo-referenced imagery navigation and VINS-based SLAM systems.
- Integrated ROS 2, PX4, MAVROS, MAVLink, and Gazebo-based simulation and autonomy workflows.
- Evaluated perception and autonomy approaches for robotics and navigation applications.

MLOps Tool Developer

Amazon Fulfillment Technologies and Robotics (Contractor) | May 2023 – Dec 2023

- Developed dataset conversion tools supporting Ambarella vision processor integration.
- Converted custom datasets for YOLO training workflows in Amazon SageMaker.
- Integrated conversion and training pipelines into Jupyter notebook environments.
- Supported machine vision model training and deployment workflows.

Machine Vision Engineer

Force Consulting Services LLC for United States Postal Service | Nov 2020 – May 2023

- Developed AI and computer vision algorithms for logistics and automation applications.
- Implemented RNN event processing for infrared range camera systems.
- Evaluated ORB-SLAM3 for facility navigation and robotic guidance.
- Developed machine vision systems for container fullness measurement and automated mail sorting.
- Built perception pipelines using Python, OpenCV, Open3D, asyncio, and related tooling.

Director of Imaging Systems

Applied Signals Intelligence Inc. | May 2019 – Nov 2020

- Developed UAV vision-based pose estimation software using ground fiducials.
- Created software interfaces for camera systems and tracking hardware.
- Developed image processing applications on Raspberry Pi platforms.
- Supported passive optical tracking and navigation applications.

President & Principal Engineer

Force Labs Inc. | Feb 2012 – Aug 2019

- Led consulting, prototyping, and multidisciplinary engineering projects for government and industry customers.
- Designed embedded systems, PCB-based electronics, software systems, and integrated architectures.
- Managed technical proposals, engineering teams, schedules, and customer coordination.
- Developed systems spanning embedded controls, sensors, electromechanical systems, and advanced technical hardware.

Founder

Iron Goat Technology Inc. | Feb 2013 – May 2016

- Pioneered biomass-powered mobile robotics systems.
- Developed stereo vision guidance systems using ROS, OpenCV, and embedded computing platforms.
- Designed brushless motor control systems and embedded electronics.
- Led system-level architecture for robotics, communications, sensing, and mechanical systems.

Education

M.S., Electrical Engineering — George Mason University, Fairfax, VA (2017)

B.S., Electrical Engineering — George Mason University, Fairfax, VA (1995)

Professional Development

Selected coursework in computer vision, AI, robotics, and related technologies.